

GED Math Practice Test 8

PART A — QUESTIONS

GED-MATH8-001

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Non-Calculator

Prompt:

A warehouse shelf holds 8 bins. Each bin contains 15 small parts. If 25 parts are removed, how many parts remain on the shelf?

- A) 95
- B) 105
- C) 120
- D) 145

GED-MATH8-002

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Non-Calculator

Prompt:

What is the value of  $4(2n + 1) - 3n$  when  $n = 6$ ?

- A) 22
- B) 28
- C) 34
- D) 52

GED-MATH8-003

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Non-Calculator

Prompt:

A bottle contains  $\frac{3}{5}$  liter of juice. A student drinks  $\frac{1}{2}$  of the juice. How many liters of juice does the student drink?

- A)  $\frac{1}{10}$
- B)  $\frac{3}{10}$
- C)  $\frac{1}{2}$
- D)  $\frac{6}{5}$

GED-MATH8-004

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Non-Calculator

Prompt:

Which expression is equivalent to  $9 - 2(3x - 5)$ ?

- A)  $-6x - 1$
- B)  $-6x + 19$

- C)  $6x - 1$   
D)  $6x + 14$

GED-MATH8-005

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Non-Calculator

Prompt:

A container holds 2.5 gallons of cleaner. How many pints of cleaner does the container hold?

- A) 5 pints  
B) 10 pints  
C) 16 pints  
D) 20 pints

GED-MATH8-006

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The formula  $A = 4s^2$  gives the area  $A$  of one side of a square sign with side length  $s$ . If  $A = 196$ , what is  $s$ ?

- A) 7  
B) 14  
C) 28  
D) 49

GED-MATH8-007

Section: Math

Type: numeric\_entry

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A box has a mass of 18.6 kilograms. Another box has a mass of 7,450 grams. What is the combined mass of the two boxes in kilograms?

Type your answer as a decimal rounded to the nearest hundredth.

GED-MATH8-008

Section: Math

Type: dropdown

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The formula  $V = \pi r^2 h$  gives the volume  $V$  of a cylinder. To solve for  $h$ , divide both sides by  $[\text{D1}]$ , so  $h = [\text{D2}]$ .

Dropdowns:

- D1)  $\pi r^2$  |  $\pi h$  |  $r^2 h$   
D2)  $V/(\pi r^2)$  |  $\pi r^2/V$  |  $V\pi r^2$

GED-MATH8-009

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A community class has 54 adults and 36 teens. What percent of the class is made up of teens?

- A) 36%
- B) 40%
- C) 54%
- D) 60%

GED-MATH8-010

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A value of  $y$  satisfies  $2.5y - 6 \leq 19$ . Which value of  $y$  is the greatest possible whole-number solution?

- A) 8
- B) 9
- C) 10
- D) 11

GED-MATH8-011

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A baker used 7.5 pounds of flour to make 120 rolls. At the same rate, how many pounds of flour are needed to make 200 rolls?

- A) 10.5
- B) 11.25
- C) 12.5
- D) 15

GED-MATH8-012

Section: Math

Type: multi\_select

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Select TWO answers.

Which expressions are equivalent to  $15 - 5x$ ?

- A)  $5(3 - x)$
- B)  $-5(x - 3)$
- C)  $5(x - 3)$
- D)  $-5x - 15$
- E)  $3(5 - x)$

GED-MATH8-013

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

The table shows the number of community garden plots by size.

Plot size: Small, Medium, Large

Number of plots: 18, 27, 15

If one plot is selected at random, what is the probability that it is medium or large?

A)  $\frac{1}{4}$

B)  $\frac{3}{5}$

C)  $\frac{7}{10}$

D)  $\frac{9}{10}$

GED-MATH8-014

Section: Math

Type: numeric\_entry

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The formula  $R = 1.8C + 32$  converts a temperature from degrees Celsius  $C$  to degrees Fahrenheit  $R$ .

If  $R = 77$ , what is  $C$ ?

Type your answer as an integer.

GED-MATH8-015

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A trapezoid has bases of 10 inches and 16 inches and a height of 7 inches. What is the area of the trapezoid?

Use  $A = \frac{1}{2}(b_1 + b_2)h$ .

A) 91 square inches

B) 98 square inches

C) 112 square inches

D) 182 square inches

GED-MATH8-016

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The table shows part of a linear relationship.

$x$ : 2, 6, 14, 18

$y$ : 11, 23, 47, 59

Which equation represents the relationship?

- A)  $y = 2x + 7$
- B)  $y = 3x + 5$
- C)  $y = 4x + 3$
- D)  $y = 6x - 1$

GED-MATH8-017

Section: Math

Type: multi\_select

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

Select TWO answers.

The table shows rainfall amounts for four months.

Month: April, May, June, July

Rainfall: 3.8 in, 5.1 in, 2.4 in, 4.7 in

Which statements are true?

- A) The total rainfall was 16.0 inches.
- B) The range was 2.7 inches.
- C) The mean rainfall was 4.5 inches.
- D) May had twice as much rainfall as June.
- E) The median rainfall was 4.7 inches.

GED-MATH8-018

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Which inequality represents “four less than the quotient of a number and 3 is greater than 11”?

- A)  $x/3 - 4 > 11$
- B)  $3/x - 4 > 11$
- C)  $4 - x/3 > 11$
- D)  $x/(3 - 4) > 11$

GED-MATH8-019

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A coat is sold for \$144 after a 20% discount. What was the original price of the coat?

- A) \$115.20
- B) \$172.80
- C) \$180.00
- D) \$288.00

GED-MATH8-020

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Solve for k:  $2(5k - 7) - 3(k + 4) = 44$ .

- A) 6
- B) 8
- C) 10
- D) 12

GED-MATH8-021

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A rectangular frame has outside dimensions 20 inches by 14 inches. The opening inside the frame is 16 inches by 10 inches. What is the area of the frame only?

- A) 80 square inches
- B) 120 square inches
- C) 160 square inches
- D) 280 square inches

GED-MATH8-022

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A line passes through the points  $(3, -2)$  and  $(9, 16)$ . What is the rate of change of the line?

- A) 2
- B) 3
- C) 4
- D) 6

GED-MATH8-023

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A solution contains 18 grams of salt in 240 grams of solution. What is the salt concentration as a percent of the solution?

- A) 7.5%
- B) 8%
- C) 12.5%
- D) 13.3%

GED-MATH8-024

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Function A is represented by  $y = -2x + 18$ . Function B is represented by the table.

x: 0, 3, 6, 9

y: 20, 11, 2, -7

Which statement is true?

- A) Function A decreases faster than Function B.
- B) Function B decreases faster than Function A.
- C) Both functions have the same rate of change.
- D) Function B has a positive rate of change.

GED-MATH8-025

Section: Math

Type: numeric\_entry

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A final grade is based on 60% written work, 25% projects, and 15% participation. A student scores 78 on written work, 92 on projects, and 88 on participation. What is the final grade?

Type your answer as a decimal rounded to the nearest tenth.

GED-MATH8-026

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A store sold 72 shirts. Plain shirts cost \$12 each, and printed shirts cost \$17 each. The total revenue was \$1,074. Which system can be used to find  $p$ , the number of plain shirts, and  $r$ , the number of printed shirts?

- A)  $p + r = 1,074$   
 $12p + 17r = 72$
- B)  $p + r = 72$   
 $12p + 17r = 1,074$
- C)  $12p + 17r = 72$   
 $p - r = 1,074$
- D)  $17p + 12r = 72$   
 $p + r = 1,074$

GED-MATH8-027

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A circular fountain has a radius of 4.5 feet. Using  $\pi \approx 3.14$ , what is the area of the fountain's surface?

- A) 28.26 square feet

- B) 56.52 square feet
- C) 63.59 square feet
- D) 127.17 square feet

GED-MATH8-028

Section: Math

Type: dropdown

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The equation  $6x + 3y = 24$  can be rewritten as  $y = [D1]x + [D2]$ .

Dropdowns:

D1) -2 | 2 | 6

D2) 8 | 24 | -8

GED-MATH8-029

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A data set has five numbers with a mean of 18. Four of the numbers are 12, 16, 20, and 25. What is the fifth number?

- A) 15
- B) 17
- C) 19
- D) 21

GED-MATH8-030

Section: Math

Type: multi\_select

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Select TWO answers.

A function is represented by  $f(x) = 2x^2 - 3x + 1$ . Which statements are true?

- A)  $f(0) = 1$
- B)  $f(1) = 0$
- C)  $f(2) = 3$
- D)  $f(-1) = 0$
- E)  $f(3) = 10$

GED-MATH8-031

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A rectangular tank has a length of 30 centimeters, a width of 18 centimeters, and a height of 20 centimeters. It is filled to 75% of its volume. How many cubic centimeters of water are in the tank?

- A) 2,700



- B) 5,400
- C) 8,100
- D) 10,800

GED-MATH8-032

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Solve for m:  $8 - 3(m - 5) = 2m + 18$ .

- A) 1
- B) 2
- C) 3
- D) 5

GED-MATH8-033

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A ladder is 26 feet long and reaches a point 24 feet up a wall. How far is the bottom of the ladder from the wall?

- A) 6 feet
- B) 8 feet
- C) 10 feet
- D) 14 feet

GED-MATH8-034

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A quadratic function is defined by  $q(x) = x^2 - 10x + 21$ . Which value of x makes  $q(x) = 0$ ?

- A) 2
- B) 3
- C) 10
- D) 21

GED-MATH8-035

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A pond contains 2,000 fish. A model predicts that the fish population will be  $2,000(1.04)^t$  after t years. What does 1.04 mean in this model?

- A) The population increases by 4% each year.
- B) The population decreases by 4% each year.

- C) The population starts at 104 fish.  
D) The population gains 1.04 fish each year.

GED-MATH8-036

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A map scale says 3 centimeters represents 42 kilometers. Two towns are 7.5 centimeters apart on the map. How far apart are the towns?

- A) 84 kilometers  
B) 96 kilometers  
C) 105 kilometers  
D) 126 kilometers

GED-MATH8-037

Section: Math

Type: drag\_drop

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Match each target with the correct tile.

Tiles:

- A)  $y = 6x + 9$   
B)  $y = 9x + 6$   
C)  $y = 6(x + 9)$   
D)  $y = 9(x + 6)$

Targets:

- T1) Multiply a number by 6, then add 9  
T2) Add 9 to a number, then multiply by 6  
T3) Add 6 to a number, then multiply by 9

GED-MATH8-038

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A delivery service charges by weight. The cost for 14 pounds is \$23.80, and the cost is proportional to weight. What is the cost for 9 pounds?

- A) \$12.60  
B) \$15.30  
C) \$17.20  
D) \$20.80

GED-MATH8-039

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Which ordered pair satisfies the inequality  $y \leq -x + 6$ ?

- A) (1, 8)
- B) (2, 4)
- C) (4, 4)
- D) (7, 1)

GED-MATH8-040

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A value is modeled by  $V = 80(0.75)^t$ . Which statement best describes the change in  $V$  each time  $t$  increases by 1?

- A)  $V$  decreases by 25%.
- B)  $V$  decreases by 75%.
- C)  $V$  increases by 25%.
- D)  $V$  increases by 75%.

GED-MATH8-041

Section: Math

Type: multi\_select

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

Select TWO answers.

The table shows the price of two package sizes of almonds.

Package: Small, Large

Weight: 12 oz, 20 oz

Price: \$4.20, \$6.40

Which statements are true?

- A) The small package costs \$0.35 per ounce.
- B) The large package costs \$0.32 per ounce.
- C) The small package has the lower unit price.
- D) The large package costs \$0.40 per ounce.
- E) The two packages have the same unit price.

GED-MATH8-042

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

For the function  $g(x) = -3x^2 + 2x + 5$ , what is  $g(-2)$ ?

- A) -21
- B) -11
- C) 1
- D) 13

GED-MATH8-043

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The table shows a relationship.

x: 1, 2, 3, 4

y: 2, 6, 12, 20

Which statement best describes the relationship?

- A) It is linear because y increases each time.
- B) It is linear because x increases by 1 each time.
- C) It is not linear because the changes in y are not constant.
- D) It is not linear because the y-values are greater than the x-values.

GED-MATH8-044

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Which expression represents 6 more than half the difference of a number x and 10?

- A)  $\frac{1}{2}x - 10 + 6$
- B)  $\frac{1}{2}(x - 10) + 6$
- C)  $\frac{1}{2}(x + 10) - 6$
- D)  $6(x - 10)/2$

GED-MATH8-045

Section: Math

Type: numeric\_entry

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The formula  $A = \frac{1}{2}h(b_1 + b_2)$  gives the area of a trapezoid. If  $A = 84$ ,  $h = 7$ , and  $b_1 = 9$ , what is  $b_2$ ?  
Type your answer as an integer.

GED-MATH8-046

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A garden path is made from 48 identical square stones. Each stone has a side length of 1.5 feet.  
What is the total area covered by the stones?

- A) 72 square feet
- B) 96 square feet
- C) 108 square feet
- D) 144 square feet

## PART B — ANSWER KEY + EXPLANATIONS

GED-MATH8-001 — Correct: A — Correct Answer: 95 — Explanation: The shelf starts with  $8 \times 15 = 120$  parts. After 25 are removed,  $120 - 25 = 95$  parts remain.

GED-MATH8-002 — Correct: C — Correct Answer: 34 — Explanation: Substitute  $n = 6$ :  $4(2(6) + 1) - 3(6) = 4(13) - 18 = 52 - 18 = 34$ .

GED-MATH8-003 — Correct: B — Correct Answer:  $3/10$  — Explanation: Half of  $3/5$  is  $3/5 \times 1/2 = 3/10$ .

GED-MATH8-004 — Correct: B — Correct Answer:  $-6x + 19$  — Explanation: Distribute:  $9 - 2(3x - 5) = 9 - 6x + 10 = -6x + 19$ .

GED-MATH8-005 — Correct: D — Correct Answer: 20 pints — Explanation: There are 8 pints in 1 gallon, so  $2.5$  gallons  $= 2.5 \times 8 = 20$  pints.

GED-MATH8-006 — Correct: A — Correct Answer: 7 — Explanation:  $196 = 4s^2$ , so  $49 = s^2$  and  $s = 7$ .

GED-MATH8-007 — Correct: 26.05 — Tolerance: exact — Correct Answer: 26.05 — Explanation:  $7,450$  grams  $= 7.45$  kilograms. The combined mass is  $18.6 + 7.45 = 26.05$  kilograms.

GED-MATH8-008 — Correct:  $D1=\pi r^2$ ;  $D2=V/(\pi r^2)$  — Correct Answer:  $D1=\pi r^2$ ;  $D2=V/(\pi r^2)$  — Explanation: Divide both sides of  $V = \pi r^2 h$  by  $\pi r^2$  to isolate  $h$ .

GED-MATH8-009 — Correct: B — Correct Answer: 40% — Explanation: There are 90 people total, and 36 are teens. Then  $36/90 = 0.40$ , or 40%.

GED-MATH8-010 — Correct: C — Correct Answer: 10 — Explanation: Solve  $2.5y - 6 \leq 19$ . Add 6 to get  $2.5y \leq 25$ , so  $y \leq 10$ .

GED-MATH8-011 — Correct: C — Correct Answer: 12.5 — Explanation: The flour per roll is  $7.5 \div 120 = 0.0625$  pound. For 200 rolls,  $200 \times 0.0625 = 12.5$  pounds.

GED-MATH8-012 — Correct: A,B — Correct Answer:  $5(3 - x) \mid -5(x - 3)$  — Explanation: Both expressions simplify to  $15 - 5x$ .

GED-MATH8-013 — Correct: C — Correct Answer:  $7/10$  — Explanation: There are 60 plots total. Medium or large plots total  $27 + 15 = 42$ . The probability is  $42/60 = 7/10$ .

GED-MATH8-014 — Correct: 25 — Tolerance: exact — Correct Answer: 25 — Explanation:  $77 = 1.8C + 32$ . Subtract 32 to get  $45 = 1.8C$ , so  $C = 25$ .

GED-MATH8-015 — Correct: A — Correct Answer: 91 square inches — Explanation:  $A = 1/2(10 + 16)(7) = 1/2(26)(7) = 91$  square inches.

GED-MATH8-016 — Correct: B — Correct Answer:  $y = 3x + 5$  — Explanation: When  $x$  increases by 4,  $y$  increases by 12, so the rate is 3. Substituting  $x = 2$  and  $y = 11$  gives  $y = 3x + 5$ .

GED-MATH8-017 — Correct: A,B — Correct Answer: The total rainfall was 16.0 inches. | The range was 2.7 inches. — Explanation: Total rainfall is  $3.8 + 5.1 + 2.4 + 4.7 = 16.0$  inches. The range is  $5.1 - 2.4 = 2.7$  inches.

GED-MATH8-018 — Correct: A — Correct Answer:  $x/3 - 4 > 11$  — Explanation: “The quotient of a number and 3” is  $x/3$ , and “four less than” that is  $x/3 - 4$ .

GED-MATH8-019 — Correct: C — Correct Answer: \$180.00 — Explanation: After a 20% discount, the sale price is 80% of the original price. So  $144 \div 0.80 = 180$ .

GED-MATH8-020 — Correct: C — Correct Answer: 10 — Explanation:  $2(5k - 7) - 3(k + 4) = 10k - 14 - 3k - 12 = 7k - 26$ . Then  $7k - 26 = 44$ , so  $k = 10$ .

GED-MATH8-021 — Correct: B — Correct Answer: 120 square inches — Explanation: Outside area is  $20 \times 14 = 280$ . Opening area is  $16 \times 10 = 160$ . Frame area is  $280 - 160 = 120$  square inches.

GED-MATH8-022 — Correct: B — Correct Answer: 3 — Explanation: Rate of change =  $(16 - (-2))/(9 - 3) = 18/6 = 3$ .

GED-MATH8-023 — Correct: A — Correct Answer: 7.5% — Explanation:  $18 \div 240 = 0.075$ , or 7.5%.

GED-MATH8-024 — Correct: B — Correct Answer: Function B decreases faster than Function A. — Explanation: Function A has rate  $-2$ . Function B has rate  $(11 - 20)/(3 - 0) = -3$ , so Function B decreases faster.

GED-MATH8-025 — Correct: 83.0 — Tolerance: exact — Correct Answer: 83.0 — Explanation: Final grade =  $0.60(78) + 0.25(92) + 0.15(88) = 46.8 + 23.0 + 13.2 = 83.0$ .

GED-MATH8-026 — Correct: B — Correct Answer:  $p + r = 72$   
 $12p + 17r = 1,074$  — Explanation: The first equation represents the number of shirts, and the second represents total revenue.

GED-MATH8-027 — Correct: C — Correct Answer: 63.59 square feet — Explanation: Area =  $\pi r^2 = 3.14(4.5^2) = 3.14(20.25) = 63.585$ , which rounds to 63.59.

GED-MATH8-028 — Correct: D1=-2; D2=8 — Correct Answer: D1=-2; D2=8 — Explanation:  $6x + 3y = 24$  gives  $3y = -6x + 24$ , so  $y = -2x + 8$ .

GED-MATH8-029 — Correct: B — Correct Answer: 17 — Explanation: Five numbers with mean 18 must total 90. The four given numbers total 73, so the fifth number is  $90 - 73 = 17$ .

GED-MATH8-030 — Correct: A,B — Correct Answer:  $f(0) = 1$  |  $f(1) = 0$  — Explanation:  $f(0) = 1$ , and  $f(1) = 2 - 3 + 1 = 0$ .

GED-MATH8-031 — Correct: C — Correct Answer: 8,100 — Explanation: Full volume is  $30 \times 18 \times 20 = 10,800$  cubic centimeters. Seventy-five percent of that is  $0.75 \times 10,800 = 8,100$ .

GED-MATH8-032 — Correct: A — Correct Answer: 1 — Explanation:  $8 - 3(m - 5) = 8 - 3m + 15 = 23 - 3m$ . Then  $23 - 3m = 2m + 18$ , so  $5 = 5m$  and  $m = 1$ .

GED-MATH8-033 — Correct: C — Correct Answer: 10 feet — Explanation: The distance from the wall is  $\sqrt{(26^2 - 24^2)} = \sqrt{(676 - 576)} = \sqrt{100} = 10$ .

GED-MATH8-034 — Correct: B — Correct Answer: 3 — Explanation:  $q(x) = x^2 - 10x + 21$  factors as  $(x - 3)(x - 7)$ , so  $x = 3$  makes  $q(x) = 0$ .

GED-MATH8-035 — Correct: A — Correct Answer: The population increases by 4% each year. — Explanation: Multiplying by 1.04 each year means the population keeps 100% and increases by 4%.

GED-MATH8-036 — Correct: C — Correct Answer: 105 kilometers — Explanation: The scale is  $42 \div 3 = 14$  kilometers per centimeter. Then  $7.5 \times 14 = 105$  kilometers.

GED-MATH8-037 — Correct: T1=A; T2=C; T3=D — Correct Answer: T1= $y = 6x + 9$ ; T2= $y = 6(x + 9)$ ; T3= $y = 9(x + 6)$  — Explanation: Parentheses show which addition happens before multiplication.

GED-MATH8-038 — Correct: B — Correct Answer: \$15.30 — Explanation: The unit price is  $23.80 \div 14 = \$1.70$  per pound. For 9 pounds, the cost is  $9 \times 1.70 = \$15.30$ .

GED-MATH8-039 — Correct: B — Correct Answer: (2, 4) — Explanation: Substitute (2, 4):  $-2 + 6 = 4$ , and  $4 \leq 4$  is true.

GED-MATH8-040 — Correct: A — Correct Answer: V decreases by 25%. — Explanation: Multiplying by 0.75 means 75% of the value remains, so the decrease is 25%.

GED-MATH8-041 — Correct: A,B — Correct Answer: The small package costs \$0.35 per ounce. | The large package costs \$0.32 per ounce. — Explanation: Small unit price:  $4.20 \div 12 = \$0.35$  per ounce. Large unit price:  $6.40 \div 20 = \$0.32$  per ounce.

GED-MATH8-042 — Correct: B — Correct Answer: -11 — Explanation:  $g(-2) = -3(-2)^2 + 2(-2) + 5 = -12 - 4 + 5 = -11$ .

GED-MATH8-043 — Correct: C — Correct Answer: It is not linear because the changes in y are not constant. — Explanation: The y-values increase by 4, then 6, then 8, so the rate of change is not constant.

GED-MATH8-044 — Correct: B — Correct Answer:  $\frac{1}{2}(x - 10) + 6$  — Explanation: “The difference of x and 10” is  $x - 10$ . Half that difference is  $\frac{1}{2}(x - 10)$ , and 6 more is  $\frac{1}{2}(x - 10) + 6$ .

GED-MATH8-045 — Correct: 15 — Tolerance: exact — Correct Answer: 15 — Explanation:  $84 = \frac{1}{2}(7)(9 + b_2)$ , so  $84 = 3.5(9 + b_2)$ . Then  $24 = 9 + b_2$ , so  $b_2 = 15$ .

GED-MATH8-046 — Correct: C — Correct Answer: 108 square feet — Explanation: Each square stone has area  $1.5 \times 1.5 = 2.25$  square feet. The total area is  $48 \times 2.25 = 108$  square feet.